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DEPARTMENT OF STATISTICS



DISSERTATION

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SIGNED DECLARATION

I affirm, that I have identified all my sources,
and that no part of my dissertation paper,
uses unacknowledged materials.

ACKNOWLEDGEMENT

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SYNOPSIS

INTRODUCTION:

- **Realization:**
I, Sulagna Roy, a UG 3rd year Final Semester 6 student of the St. Xavier's College (Autonomous), Kolkata, realized during these 2 years of my college life, that the UG and PG students of the College, on an average or rather mostly, are not happy with their obtained Semester Marks.
- **Decision:**
I aimed at analyzing the most necessary and common factors, influencing a UG or a PG student's Semester Marks, under the consideration of focusing only on the even Semesters, because these are basically the last semesters of an Academic Year of any Course concerned, that categorize a student as either 'Promoted to the next Academic Year', or, as 'Passed out of the College Successfully'.

➤ POPULATION:

- **Nature:** College Students currently pursuing their *Bachelors or, Masters*.
- **Geographical Location Concerned:** Students of the *St. Xavier's College (Autonomous), Kolkata*.
- **Time Period Concerned:**
 - ✓ Semester No. - An *Even* Semester i.e.,
 - i. *2 or 4 or 6* under the Under-Graduate (*UG*) Courses.
 - ii. *2 or 4* under the Post-Graduate (*PG*) Courses.
 - ✓ Common Academic Period of the Even Semester – *January,2024 to June,2024*.

➤ POPULATION RESPONSE VARIABLE:

Let, Y: A student's *obtained percentage, out of the total marks allotted*, for the concerned course, in the concerned Even Semester, w.r.t. the student being selected from the previously mentioned population.

➤ POPULATION PREDICTORS:

We have selected *5 predictors*, keeping in mind how students from the stated population, prepared themselves for their aimed percentage marks out of the total marks allotted, for the concerned course, in the concerned Even Semesters.

The predictors, with respect to a student concerned, are-

1. X₁: Daily Average Duration (in hours) of *Self-Study*
2. X₂: Daily Average Duration (in hours) of *Physical Exercise*
3. X₃: Daily Average Duration (in hours) of *Sleep*
4. X₄: Daily Average Duration (in hours) of *Travel Time*
5. X₅: Daily Average Duration (in hours) of *Media-usage*

➤ **STATISTICAL TOPICS & METHODOLOGY & CONCLUSIONS:**

1. *Sample survey* for Sample size selection and sampling technique.
2. *Descriptive Analysis* of the Sample Marks Collected.
3. *Bivariate Analysis* for each of the predictors with the response.
4. *Multivariate Analysis* combining all the predictors and response.

DISSERTATION

➤ **IMPORTANT RESTRICTIONS & RECONSIDERATIONS IN THE DISSERTATION WORK:**

Note that all the restrictions have been considered for easy data collection as per time constraint.

1. Population Time Period-

Here instead of considering the whole population of the UG and the PG students, we consider only the current UG Semester 6 students of 2025 of only Statistics Department, who were previously from the UG Semester 4 as of 2024.

1. Reasons-

i. Why only Semester 4 from UG?

There are chances, that Students from the even Semesters, other than that of Semester 6 may not be a part of the college currently, so we exclude those UG students who were previously of Semesters 2 or 6 and are accordingly of Semester 4 under UG or of PG semester 2, respectively.

ii. Why not PG students?

Similarly, the PG students who were previously in Semester 4 have been excluded.

iii. Why only Statistics Department?

In order to consider a Homogeneous Population as per Marks we have considered the Science Department, because of the usual realization that- students from the Arts Department on an average get lesser marks, while students from the Commerce Department on an average get higher marks, than the Students from the Science Department, and for time constraint and easy data collection purpose a particular case i.e., the Statistics Department has been considered.

2. Population Nature as per marks obtained-

Here, we would consider only those students who *didn't have any Arrear Papers*.

2. Reasons-

i. Direct Result Publication-

The marks of these students get published on direct publication of the results.

ii. True Response-

In order to maintain the comfort level of the respondents while answering the Questionnaire (to be mentioned explicitly later), the question regarding marks has intentionally being excluded, and the students in the sample will be identified by their roll numbers.

METHODOLOGY

➤ STAGE 1---POPULATION SETUP

■ POPULATION-

- a. **Geographical Location**-Students of the St. Xavier's College (Autonomous), Kolkata.
- b. **Department**-Statistics (UG)
- c. **Semester & Academic Session Details**-
 - ✓ *Current Even Semester no.:* 6
Current Academic Session: January,2025 to June,2025
 - ✓ *Previous Even Semester no.:* 4
Current Academic Session: January,2024 to June,2024.

■ POPULATION STUDY VARIALES-

In total we have $p=6$ population variables w.r.t. a student from the population, on which sample data is collected-

1. X_1 : Obtained *Percentage marks* out of full marks of 500 in Semester 4
2. X_2 : Daily Average Percentage Duration of *Self-Study*
3. X_3 : Daily Average Percentage Duration of *Physical Exercise*
4. X_4 : Daily Average Percentage Duration of *Sleep*
5. X_5 : Daily Average Percentage Duration of *Travel Time*
6. X_6 : Daily Average Percentage Duration of *Media-usage*

• NOTE-

Here, obviously except X_1 is the response and all other 5 are the predictors.

■ POPULATION SIZE-

As per the restrictions concerned a total of $N=39$ students have passed all the papers successfully, so here our population size is $N=39$.

➤ STAGE 2---QUESTIONNAIRE

• QUESTIONS-

Only *6 questions* were being asked in total-

1. Roll no.:(3 digitd) [MCQ Format]
2. Daily Average Durations[(hr(s): min(s): sec(s)) Format] of the following-
 - i. Self-Study
 - ii. Physical Exercise
 - iii. Sleep
 - iv. Travel
 - v. Media- Usage
 - vi.

• LINK-

<https://forms.gle/z8bEMJLpSonmj9Rz5>

• SCREENSHOTS-

INFLUENCE OF DAILY ROUTINE ON SEMESTER MARKS

- ***Thank you*** for choosing to answer this Online Questionnaire on such an ***Important topic of our College Life.***
- All the questions concerned do not reveal your ***personal identity.***
- So, feel free to ***answer honestly.***
- It would just take a ***few seconds.***

sulagna1roy@gmail.com [Switch account](#)
Not shared

* Indicates required question

Roll no. : (3 digit) *

☐ 469
☐ 409
☐ 415
☐ 406
☐ 445
☐ 462
☐ 458

☐ 462
☐ 458
☐ 418
☐ 421
☐ 422
☐ 427
☐ 472
☐ 467
☐ 456
☐ 411
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☐ 428
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☐ 464
☐ 425
☐ 432
☐ 465
☐ 402
☐ 442
☐ 468
☐ 451
☐ 461
☐ 470
☐ 466
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☐ 449
☐ 410
☐ 414
☐ 429
☐ 426
☐ 416
☐ 431
☐ 450
☐ 420
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☐ 435
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☐ 403
☐ 448
☐ 413
☐ 404
☐ 441
☐ 408
☐ 444
☐ 446
☐ 443

Next

This co

Media Usage : *

Hrs Min Sec

: :

Back Submit

This content is neither cre

Sleep : *

Hrs Min Sec

: :

Travel Time : *

Hrs Min Sec

: :

DAILY ROUTINE

(Answer the following as in terms of 'Daily Average Duration')

Self Study : *

Hrs Min Sec

: :

> STAGE 3---FIXED SAMPLE

Only $n=25$ students have answered so that is our fixed sample of size $n=25$.

➤ STAGE 4---DATA COLLECTION & DATA PROCESSING

1. STEP 1-TABLE NO. 1: COLLECTED DATA SET:

Using 'Google Forms' we were directly being able to form the Excel Sheet of the 'Collected Data Set' that is what has been presented through this table beautifully.

TABLE NO. 1: COLLECTED DATA SET						
SL. NO.S	ROLL NO.S	DAILY AVERAGE DURATIONS (hr(s):min(s):sec(s))				
		SELF-STUDY	PHYSICAL EXERCISE	SLEEP	TRAVEL	MEDIA USAGE
1	404	01:30:00	00:30:00	07:00:00	02:30:00	04:00:00
2	406	03:00:00	00:20:00	07:00:00	02:00:00	02:10:00
3	408	04:30:00	00:00:00	07:00:00	02:00:00	02:00:00
4	410	02:00:00	00:00:00	05:00:00	04:00:00	02:00:00
5	411	02:00:00	01:00:00	07:00:00	02:00:00	02:00:00
6	413	03:00:00	00:10:00	06:00:00	00:00:00	03:00:00
7	414	03:00:00	00:45:00	07:00:00	04:00:00	01:30:00
8	415	03:00:00	01:00:00	05:30:00	00:20:00	03:00:00
9	416	02:00:00	00:10:00	08:30:00	00:10:00	02:30:00
10	418	05:00:00	00:30:00	06:00:00	03:00:00	02:30:00
11	420	02:30:00	00:30:00	06:00:00	02:00:00	05:00:00
12	421	04:00:00	01:00:00	08:00:00	00:30:00	04:00:00
13	422	02:00:00	00:30:00	07:00:00	00:30:00	01:00:00
14	428	07:00:00	00:30:00	08:00:00	00:15:00	01:00:00
15	432	03:00:00	01:00:00	08:00:00	02:00:00	05:00:00
16	441	04:30:15	00:00:00	07:00:00	00:35:00	04:00:00
17	442	01:00:00	00:30:00	07:00:00	02:00:00	01:00:00
18	444	03:00:00	02:00:00	06:00:00	00:40:00	04:00:00
19	445	03:00:00	00:30:00	06:00:00	04:00:00	03:00:00
20	448	02:30:00	00:15:00	07:30:00	02:00:00	03:00:00
21	450	03:00:00	01:30:00	07:00:00	03:00:00	03:30:00
22	458	04:30:00	01:00:00	07:30:00	01:00:00	01:00:00
23	465	02:00:00	00:15:00	07:00:00	03:00:00	04:00:00
24	466	01:00:00	00:00:00	07:00:00	02:30:00	02:00:00
25	468	00:30:00	00:00:00	11:00:00	01:00:00	05:00:00

2. STEP 2-TABLE NO. 2: PROCESSED DATA SET 1:

Here from the previous 'TABLE NO. 1', each cell value for each of the 5 predictors which were previously in 'hr(s): min(s): sec(s)' format have been converted to 'sec(s)', and a new column for 'Semester 4 Marks' i.e., in short 'Sem 4 Marks' has been included. The calculations were done in 'Offline Mode' by 'pen-paper-pencil.'

TABLE NO. 2: PROCESSED DATA SET 1							
SL. NO.S	ROLL NO.S	SEM 4 MARKS (/500)	DAILY AVERAGE DURATIONS (sec(s))				
			SELF-STUDY	PHYSICAL EXERCISE	SLEEP	TRAVEL	MEDIA USAGE
1	404	390	5400	1800	25200	9000	14400
2	406	412	10800	1200	25200	7200	7800
3	408	412	16200	0	25200	7200	7200
4	410	313	7200	0	18000	14400	7200
5	411	352	7200	3600	25200	7200	7200
6	413	342	10800	6000	21600	0	10800
7	414	376	10800	2700	25200	14400	5400
8	415	328	10800	3600	19800	1200	10800
9	416	347	7200	6000	30600	600	9000
10	418	447	18000	1800	21600	10800	9000
11	420	392	9000	1800	21600	7200	18000
12	421	302	14400	3600	28800	1800	14400
13	422	325	7200	1800	25200	1800	3600
14	428	318	25200	1800	28800	900	3600
15	432	344	10800	3600	28800	7200	18000
16	441	369	16215	0	25200	2100	14400
17	442	439	3600	1800	25200	7200	3600
18	444	292	10800	7200	21600	2400	14400
19	445	419	10800	1800	21600	14400	10800
20	448	455	9000	900	27000	7200	10800
21	450	415	10800	5400	25200	10800	12600
22	458	329	16200	3600	27000	3600	3600
23	465	335	7200	900	25200	10800	14400
24	466	346	3600	0	25200	9000	7200
25	468	450	1800	0	39600	3600	18000

3. STEP 3-TABLE NO. 3: PROCESSED DATA SET 2:

Here from the previous 'TABLE NO. 2', each cell value for each of the 6 variables have been converted to values belonging to a Closed Interval of [0,1] by dividing each value in the 'Marks' column by 500 and by dividing each of the other 5 columns which by the total secs in a day of 24(hrs)*60(mins)*60(secs)=86400 (secs) using 'Minitab'.

TABLE NO. 3 : PROCESSED DATA SET 2							
SL. NO.S	ROLL NO.S	SEM 4 MARKS ([0,1])	DAILY AVERAGE DURATIONS ([0,1])				
			SELF-STUDY	PHYSICAL EXERCISE	SLEEP	TRAVEL	MEDIA USAGE
1	404	0.78	0.0625	0.020833	0.291667	0.104167	0.1666667
2	406	0.824	0.125	0.013889	0.291667	0.083333	0.0902778
3	408	0.824	0.1875	0	0.291667	0.083333	0.0833333
4	410	0.626	0.0833333	0	0.208333	0.166667	0.0833333
5	411	0.704	0.0833333	0.041667	0.291667	0.083333	0.0833333
6	413	0.684	0.125	0.069444	0.25	0	0.125
7	414	0.752	0.125	0.03125	0.291667	0.166667	0.0625
8	415	0.656	0.125	0.041667	0.229167	0.013889	0.125
9	416	0.694	0.0833333	0.069444	0.354167	0.006944	0.1041667
10	418	0.894	0.2083333	0.020833	0.25	0.125	0.1041667
11	420	0.784	0.1041667	0.020833	0.25	0.083333	0.2083333
12	421	0.604	0.1666667	0.041667	0.333333	0.020833	0.1666667
13	422	0.65	0.0833333	0.020833	0.291667	0.020833	0.0416667
14	428	0.636	0.2916667	0.020833	0.333333	0.010417	0.0416667
15	432	0.688	0.125	0.041667	0.333333	0.083333	0.2083333
16	441	0.738	0.1876736	0	0.291667	0.024306	0.1666667
17	442	0.878	0.0416667	0.020833	0.291667	0.083333	0.0416667
18	444	0.584	0.125	0.083333	0.25	0.027778	0.1666667
19	445	0.838	0.125	0.020833	0.25	0.166667	0.125
20	448	0.91	0.1041667	0.010417	0.3125	0.083333	0.125
21	450	0.83	0.125	0.0625	0.291667	0.125	0.1458333
22	458	0.658	0.1875	0.041667	0.3125	0.041667	0.0416667
23	465	0.67	0.0833333	0.010417	0.291667	0.125	0.1666667
24	466	0.692	0.0416667	0	0.291667	0.104167	0.0833333
25	468	0.9	0.0208333	0	0.458333	0.041667	0.2083333

4. STEP 4-TABLE NO. 4: PROCESSED DATA SET 3:

Here from the previous 'TABLE NO. 3', each cell value for each of the 6 variables were being converted to values belonging to the Closed Interval of [0,100] i.e., as in % values to match with the scale of measurement of the defined 6 variables concerned, by just multiplying each of the 6 columns concerned by 100, using 'Minitab'.

TABLE NO. 4 : PROCESSED DATA SET 3							
SL. NO.S	ROLL NO.S	SEM 4 % MARKS ([0,100])	DAILY AVERAGE % DURATIONS ([0,100])				
			SELF-STUDY	PHYSICAL EXERCISE	SLEEP	TRAVEL	MEDIA USAGE
1	404	78	6.25	2.083333	29.16667	10.41667	16.666667
2	406	82.4	12.5	1.388889	29.16667	8.333333	9.0277778
3	408	82.4	18.75	0	29.16667	8.333333	8.3333333
4	410	62.6	8.3333333	0	20.83333	16.66667	8.3333333
5	411	70.4	8.3333333	4.166667	29.16667	8.333333	8.3333333
6	413	68.4	12.5	6.944444	25	0	12.5
7	414	75.2	12.5	3.125	29.16667	16.66667	6.25
8	415	65.6	12.5	4.166667	22.91667	1.388889	12.5
9	416	69.4	8.3333333	6.944444	35.41667	0.694444	10.416667
10	418	89.4	20.833333	2.083333	25	12.5	10.416667
11	420	78.4	10.416667	2.083333	25	8.333333	20.833333
12	421	60.4	16.666667	4.166667	33.33333	2.083333	16.666667
13	422	65	8.3333333	2.083333	29.16667	2.083333	4.1666667
14	428	63.6	29.166667	2.083333	33.33333	1.041667	4.1666667
15	432	68.8	12.5	4.166667	33.33333	8.333333	20.833333
16	441	73.8	18.767361	0	29.16667	2.430556	16.666667
17	442	87.8	4.1666667	2.083333	29.16667	8.333333	4.1666667
18	444	58.4	12.5	8.333333	25	2.777778	16.666667
19	445	83.8	12.5	2.083333	25	16.66667	12.5
20	448	91	10.416667	1.041667	31.25	8.333333	12.5
21	450	83	12.5	6.25	29.16667	12.5	14.583333
22	458	65.8	18.75	4.166667	31.25	4.166667	4.1666667
23	465	67	8.3333333	1.041667	29.16667	12.5	16.666667
24	466	69.2	4.1666667	0	29.16667	10.41667	8.3333333
25	468	90	2.0833333	0	45.83333	4.166667	20.833333

5. STEP 5-TABLE NO. 5: PROCESSED DATA SET 4:

Here from the previous 'TABLE NO. 4', for easy calculation purposes we have approximated each cell value of the 6 columns concerned by rounding off the values to 2 decimal places, using 'Minitab'.

TABLE NO. 5 : PROCESSED DATA SET 4							
SL. NO.S	ROLL NO.S	SEM 4 % MARKS ([0,100])	DAILY AVERAGE % DURATIONS ([0,100], rounded off)				
			SELF-STUDY	PHYSICAL EXERCISE	SLEEP	TRAVEL	MEDIA USAGE
1	404	78	6.25	2.08	29.17	10.42	16.67
2	406	82.4	12.5	1.39	29.17	8.33	9.03
3	408	82.4	18.75	0	29.17	8.33	8.33
4	410	62.6	8.33	0	20.83	16.67	8.33
5	411	70.4	8.33	4.17	29.17	8.33	8.33
6	413	68.4	12.5	6.94	25	0	12.5
7	414	75.2	12.5	3.13	29.17	16.67	6.25
8	415	65.6	12.5	4.17	22.92	1.39	12.5
9	416	69.4	8.33	6.94	35.42	0.69	10.42
10	418	89.4	20.83	2.08	25	12.5	10.42
11	420	78.4	10.42	2.08	25	8.33	20.83
12	421	60.4	16.67	4.17	33.33	2.08	16.67
13	422	65	8.33	2.08	29.17	2.08	4.17
14	428	63.6	29.17	2.08	33.33	1.04	4.17
15	432	68.8	12.5	4.17	33.33	8.33	20.83
16	441	73.8	18.77	0	29.17	2.43	16.67
17	442	87.8	4.17	2.08	29.17	8.33	4.17
18	444	58.4	12.5	8.33	25	2.78	16.67
19	445	83.8	12.5	2.08	25	16.67	12.5
20	448	91	10.42	1.04	31.25	8.33	12.5
21	450	83	12.5	6.25	29.17	12.5	14.58
22	458	65.8	18.75	4.17	31.25	4.17	4.17
23	465	67	8.33	1.04	29.17	12.5	16.67
24	466	69.2	4.17	0	29.17	10.42	8.33
25	468	90	2.08	0	45.83	4.17	20.83

6. STEP 6-TABLE NO. 6: WORKING DATA SET:

Here from the previous 'TABLE NO. 5', we have just changed the column names by the usual well defined observed sample variable notations based on the usual well defined population variable notations.

TABLE NO. 6 : WORKING DATA SET						
i	x1i	x2i	x3i	x4i	x5i	x6i
1	78	6.25	2.08	29.17	10.42	16.67
2	82.4	12.5	1.39	29.17	8.33	9.03
3	82.4	18.75	0	29.17	8.33	8.33
4	62.6	8.33	0	20.83	16.67	8.33
5	70.4	8.33	4.17	29.17	8.33	8.33
6	68.4	12.5	6.94	25	0	12.5
7	75.2	12.5	3.13	29.17	16.67	6.25
8	65.6	12.5	4.17	22.92	1.39	12.5
9	69.4	8.33	6.94	35.42	0.69	10.42
10	89.4	20.83	2.08	25	12.5	10.42
11	78.4	10.42	2.08	25	8.33	20.83
12	60.4	16.67	4.17	33.33	2.08	16.67
13	65	8.33	2.08	29.17	2.08	4.17
14	63.6	29.17	2.08	33.33	1.04	4.17
15	68.8	12.5	4.17	33.33	8.33	20.83
16	73.8	18.77	0	29.17	2.43	16.67
17	87.8	4.17	2.08	29.17	8.33	4.17
18	58.4	12.5	8.33	25	2.78	16.67
19	83.8	12.5	2.08	25	16.67	12.5
20	91	10.42	1.04	31.25	8.33	12.5
21	83	12.5	6.25	29.17	12.5	14.58
22	65.8	18.75	4.17	31.25	4.17	4.17
23	67	8.33	1.04	29.17	12.5	16.67
24	69.2	4.17	0	29.17	10.42	8.33
25	90	2.08	0	45.83	4.17	20.83

➤ STAGE 5---DESCRIPTIVE STATISTICAL ANALYSIS & ESTIMATION**I. R CODE--**###STEP 1---REMOVE MEMORY

```
rm=(list=ls())
```

###STEP 2---->DATA INPUTS##STEP 2.1---->Sample Size:

```
n=25
```

##STEP 2.2---->Total no. of variables:

```
p=6
```

##STEP 2.3---->Observed 6 vectors:*#STEP 2.3.1---->%Marks (Response/variable 1)*

```
x1=c(78,82.4,82.4,62.6,70.4,68.4,75.2,65.6,69.4,89.4,78.4,60.4,  
65,63.6,68.8,73.8,87.8,58.4,83.8,91,83,65.8,67,69.2,90)
```

#STEP 2.3.2---->%Self-Study (Predictor 1/variable 2)

```
x2=c(6.25,12.5,18.75,8.33,8.33,12.5,12.5,12.5,8.33,20.83,10.42,16.67,  
8.33,29.17,12.5,18.77,4.17,12.5,12.5,10.42,12.5,18.75,8.33,4.17,2.08)
```

#STEP 2.3.3---->%Physical Exercise (Predictor 2/variable 3)

```
x3=c(2.08,1.39,0,0,4.17,6.94,3.13,4.17,6.94,2.08,2.08,4.17,2.08,  
2.08,4.17,0,2.08,8.33,2.08,1.04,6.25,4.17,1.04,0,0)
```

#STEP 2.3.4---->%Sleep (Predictor 3/variable 4)

```
x4=c(29.17,29.17,29.17,20.83,29.17,25,29.17,22.92,35.42,25,25,33.33,  
29.17,33.33,33.33,29.17,29.17,25,25,31.25,29.17,31.25,29.17,29.17,45.83)
```

#STEP 2.3.5---->%Travel (Predictor 4/variable 5)

```
x5=c(10.42,8.33,8.33,16.67,8.33,0,16.67,1.39,0.69,12.5,8.33,2.08,2.08,1.04,  
8.33,2.43,8.33,2.78,16.67,8.33,12.5,4.17,12.5,10.42,4.17)
```

#STEP 2.3.6---->%Media-Usage (Predictor 5/variable 6)

```
x6=c(16.67,9.03,8.33,8.33,8.33,12.5,6.25,12.5,10.42,10.42,20.83,16.67,4.17,  
4.17,20.83,16.67,4.17,16.67,12.5,12.5,14.58,4.17,16.67,8.33,20.83)
```

###STEP 3---->1-6-VARIATES LINEAR REGRESSION MODEL ESTIMATION

##STEP 3.1---->Data Matrix-

```
X=matrix(c(x1,x2,x3,x4,x5,x6),byrow=F,ncol=6,nrow=25)
```

```
X
```

##STEP 3.2---->Mean Vector-

```
xb=apply(X,2,mean)
```

```
xb
```

##STEP 3.3---->Dispersion Matrix-

```
S=matrix(0,nrow=p,ncol=p)
```

```
for(j in 1:p)
```

```
{
```

```
  for(k in 1:p)
```

```
  {
```

```
    S[j,k]=(sum((X[,j]-xb[j])*(X[,k]-xb[k])))/n
```

```
  }
```

```
}
```

##STEP 3.4---->Application of Cramer's Rule-

##STEP 3.4.1---->Modified 'D' Matrix-

```
D=matrix(0,nrow=p-1,ncol=p-1)
```

```
for(i in 1:p-1)
```

```
{
```

```
  for(j in 1:p-1)
```

```
  {
```

```
    D[i,j]=S[i+1,j+1]
```

```
  }
```

```
}
```

##STEP 3.4.2---->Applicability Condition of Cramer's Rule:

```
library(Matrix)
```

```
rankMatrix(D)
```

##STEP 3.4.3---->Modified 'D1' Vector-

```
D1=matrix(0,nrow=p-1,ncol=1)
```

```
for(i in 1:p-1)
```

```
{
```

```
  D1[i]=S[i+1,1]
```

```
}
```

4

2

#STEP 3.4.4---->Regression Coefficients Estimates-

#STEP 3.4.4.1---->

```
D2=matrix(c(D1,D[,2],D[,3],D[,4],D[,5]),byrow=F,nrow=p-1,ncol=p-1,)
```

```
D2
```

```
b2=det(D2)/det(D)
```

```
b2
```

#STEP 3.4.4.2---->

```
D3=matrix(c(D[,1],D1,D[,3],D[,4],D[,5]),byrow=F,nrow=p-1,ncol=p-1,)
```

```
D3
```

```
b3=det(D3)/det(D)
```

```
b3
```

#STEP 3.4.4.3---->

```
D4=matrix(c(D[,1],D[,2],D1,D[,4],D[,5]),byrow=F,nrow=p-1,ncol=p-1,)
```

```
D4
```

```
b4=det(D4)/det(D)
```

```
b4
```

#STEP 3.4.4.4---->

```
D5=matrix(c(D[,1],D[,2],D[,3],D1,D[,5]),byrow=F,nrow=p-1,ncol=p-1,)
```

```
D5
```

```
b5=det(D5)/det(D)
```

```
b5
```

#STEP 3.4.4.5---->

```
D6=matrix(c(D[,1],D[,2],D[,3],D[,4],D1),byrow=F,nrow=p-1,ncol=p-1,)
```

```
D6
```

```
b6=det(D6)/det(D)
```

```
b6
```

#STEP 3.4.4.6---->

```
b1=(xb[1])-(b2*xb[2])-(b3*xb[3])-(b4*xb[4])-(b5*xb[5])
```

```
b1
```

###STEP 4---->5-'2-VARIATES' LINEAR REGRESSION MODELS ESTIMATION

##STEP 4.1---->2-Variate Linear Model no.1 %Marks: %Self-Study-

```
X1_2=lm(x1~x2)
```

##STEP 4.2---->2-Variate Linear Model no.2 %Marks: %Physical Exercise-

```
X1_3=lm(x1~x3)
```

##STEP 4.3---->2-Variate Linear Model no.3 %Marks: %Sleep-

```
X1_4=lm(x1~x4)
```

##STEP 4.4---->2-Variate Linear Model no.4 %Marks: %Travel-

```
X1_5=lm(x1~x5)
```

##STEP 4.5---->2-Variate Linear Model no.5 %Marks: %Media-Usage-

```
X1_6=lm(x1~x6)
```

###STEP 5---->6-'1-VARIATE' ANALYSIS

##STEP 5.1---->Central Tendency Measure—Arithmetic Mean values

```
rm1=xb
```

```
rm1
```

##STEP 5.2---->Dispersion Measure--Standard Deviation Values

```
cm2=matrix(0,nrow=p,ncol=1)
```

```
for(i in 1:p)
```

```
{
```

```
  cm2[i]=S[i,i]
```

```
}
```

```
cm2
```

```
sd=cm2^(0.5)
```

```
sd
```

##STEP 5.3---->Skewness Measure--g1 values

```
cm3= matrix(0,nrow=p,ncol=1)
```

```
cm3=matrix(0,nrow=p,ncol=1)
```

```
for(i in 1:p)
```

```
{
```

```
  cm3[i]=(sum((X[,i]-rm1[i])^3))/n
```

```
}
```

```
cm3
```

```
g1=cm3/(sd^3)
```

```
g1
```

##STEP 5.4---->Kurtosis Measure--g2 values

```
cm4= matrix(0,nrow=p,ncol=1)
```

```
cm4=matrix(0,nrow=p,ncol=1)
```

```
for(i in 1:p)
```

```
{
```

```
  cm4[i]=(sum((X[,i]-rm1[i])^4))/n
```

```
}
```

```
cm4
```

```
g2=(cm4/(cm2^2))-3
```

```
g2
```

##STEP 5.5---->1 Variate Analysis Matrix

```
UANAM=matrix(c(rm1,sd,g1,g2),byrow=T,ncol=p,nrow=4)
```

```
rownames(UANAM)=c('rm1','sd','g1','g2')
```

```
colnames(UANAM)=c('M','SS','PE','SL','TR','MU')
```

```
UANAM
```

II. ANSWERS & INTERPRETATIONS-

Here we would like to introduce the 6 short-forms of the 6 Variables concerned for easy interpretation of our obtained conclusions as given in the table below-

TABLE NO. 7: SHORT-FULL FORMS' TABLE

SL. NO.S	SHORT FORMS	FULL FORMS
1	M	Obtained Percentage SEMESTER 4 MARKS out of 500
2	SS	Daily Average Percentage Duration of SELF STUDY
3	PE	Daily Average Percentage Duration of PHYSICAL EXERCISE
4	SL	Daily Average Percentage Duration of SLEEP
5	TR	Daily Average Percentage Duration of TRAVEL
6	MU	Daily Average Percentage Duration of MEDIA USAGE

➤ PART 1-----1--'6-VARIATES' LINEAR MODEL ESTIMATION

■ MODEL ESTIMATE-

$$M = 53.12845 - (0.04333403*SS) - (0.8801957*PE) + (0.6197415*SL) + (0.7585131*TR) + (0.1120001*MU)$$

■ STEP-WISE MODEL ESTIMATE'S INTERPRETATIONS-

✓ Intercept term: (+53.12845)

A student from the concerned population is estimated to obtain a *minimum percentage marks of 53.12845%*, as per the above mentioned '6-Variates Estimated Linear Regression Model'.

✓ 5 - 'multiple' regression coefficient terms-

i. Coefficient no.1: (-0.04333403)

As per the above mentioned '6-Variates Estimated Linear Regression Model', keeping the values of other predictors fixed, out of 86400 secs in a day increase by 1sec in Self-Study duration is estimated to decrease the marks by 0.04333403%, i.e., out of 14400 mins in a day increase by 1min in Self-Study duration is estimated to decrease the marks by 2.6000418(=0.04333403*60) %.

ii. Coefficient no.2: (-0.8801957)

As per the above mentioned '6-Variates Estimated Linear Regression Model', keeping the values of other predictors fixed, out of 86400 secs in a day increase by 1sec in Physical Exercise duration is estimated to decrease the marks by 0.8801957%, i.e., out of 14400 mins in a day increase by 1min in Physical Exercise duration is estimated to decrease the marks by 52.811742(=0.8801957*60) %.

iii. Coefficient no.3: (+0.6197415)

As per the above mentioned '6-Variates Estimated Linear Regression Model', keeping the values of other predictors fixed, out of 86400 secs in a day increase by 1sec in Sleep duration is estimated to increase the marks by 0.6197415%, i.e., out of 14400 mins in a day increase by 1min in Sleep duration is estimated to increase the marks by 37.18449(=0.6197415*60) %.

iv. **Coefficient no.4: (+0.7585131)**

As per the above mentioned '6-Variates Estimated Linear Regression Model', keeping the values of other predictors fixed, out of 86400 secs in a day increase by 1sec in Travel duration is estimated to increase the marks by 0.7585131%, i.e., out of 14400 mins in a day increase by 1min in Travel duration is estimated to increase the marks by $45.510786(=0.7585131*60)$ %.

v. **Coefficient no.5: (+0.1120001)**

As per the above mentioned '6-Variates Estimated Linear Regression Model', keeping the values of other predictors fixed, out of 86400 secs in a day increase by 1sec in Media-Usage duration is estimated to increase the marks by 0.1120001%, i.e., out of 14400 mins in a day increase by 1min in Media-Usage duration is estimated to increase the marks by $6.720006(=0.1120001*60)$ %.

➤ **PART 2-----5 --'2-VARIATES' LINEAR MODEL ESTIMATION-**

1. **2-VARIATES MODEL NO.1--%MARKS: %SELE-STUDY**

▪ **MODEL ESTIMATE-**

$$M=77.7113-(0.3078*SS)$$

▪ **STEP-WISE MODEL ESTIMATE'S INTERPRETATIONS-**

✓ **Intercept term: 77.7113**

A student from the concerned population is estimated to obtain a *minimum percentage marks of 77.7113%*, as per the above mentioned '2-Variates Estimated Linear Regression Model no.1'.

✓ **Regression coefficient term: (-0.3078)**

As per the above mentioned '2-Variates Estimated Linear Regression Model no.1', out of 86400 secs in a day increase by 1sec in Self-Study duration is estimated to decrease the marks by 0.3078%, i.e., out of 14400 mins in a day increase by 1min in Self-Study duration is estimated to decrease the marks by $18.468(=0.3078*60)$ %.

2. **2-VARIATES MODEL NO.2--%MARKS: %PHYSICAL EXERCISE**

▪ **MODEL ESTIMATE-**

$$M=78.561-(1.621*PE)$$

▪ **STEP-WISE MODEL ESTIMATE'S INTERPRETATIONS-**

✓ **Intercept term: 78.561**

A student from the concerned population is estimated to obtain a *minimum percentage marks of 78.561%*, as per the above mentioned '2-Variates Estimated Linear Regression Model no.2'.

✓ **Regression coefficient term: (-1.621)**

As per the above mentioned '2-Variates Estimated Linear Regression Model no.2', out of 86400 secs in a day increase by 1sec in Physical Exercise duration is estimated to decrease the marks by 1.621%, i.e., out of 14400 mins in a day increase by 1min in Physical Exercise duration is estimated to decrease the marks by $97.26(=1.621*60)$ %.

3. **2-VARIATES MODEL NO.3--%MARKS: %SLEEP**

▪ **MODEL ESTIMATE-**

$$M=61.3151+(0.4322*SL)$$

▪ **STEP-WISE MODEL ESTIMATE'S INTERPRETATIONS-**

✓ **Intercept term: 61.3151**

A student from the concerned population is estimated to obtain a *minimum percentage marks of 61.3151%*, as per the above mentioned '2-Variates Estimated Linear Regression Model no.3'.

✓ **Regression coefficient term: (+0.4322)**

As per the above mentioned '2-Variates Estimated Linear Regression Model no.3', out of 86400 secs in a day increase by 1sec in Sleep duration is estimated to increase the marks by 0.4322%, i.e., out of 14400 mins in a day increase by 1min in Sleep duration is estimated to increase the marks by $25.932(=0.4322*60)$ %.

4. 2-VARIATES MODEL NO.4--%MARKS: %TRAVEL

▪ MODEL ESTIMATE-

$$M=68.5373 + (0.7273*TR)$$

▪ STEP-WISE MODEL ESTIMATE'S INTERPRETATIONS-

✓ Intercept term: 68.5373

A student from the concerned population is estimated to obtain

a minimum percentage marks of 68.5373%,

as per the above mentioned '2-Variates Estimated Linear Regression Model no.4'.

✓ Regression coefficient term: (+0.7273)

As per the above mentioned '2-Variates Estimated Linear Regression Model no.4',

out of 86400 secs in a day increase by 1sec in Travel duration is estimated to increase the marks by 0.7273%, i.e., out of 14400 mins in a day increase by 1min in Travel duration is estimated to increase the marks by 43.638(=0.7273*60) %.

5. 2-VARIATES MODEL NO.5--%MARKS: %MEDIA-USAGE

▪ MODEL ESTIMATE-

$$M= 72.058 + (0.163 *MU)$$

▪ STEP-WISE MODEL ESTIMATE'S INTERPRETATIONS-

✓ Intercept term: 72.058

A student from the concerned population is estimated to obtain

a minimum percentage marks of 72.058%,

as per the above mentioned '2-Variates Estimated Linear Regression Model no.5'.

✓ Regression coefficient term: (+0.163)

As per the above mentioned '2-Variates Estimated Linear Regression Model no.5',

out of 86400 secs in a day increase by 1sec in Media-Usage duration is estimated to increase the marks by 0.163 %, i.e., out of 14400 mins in a day increase by 1min in Media-Usage duration is estimated to increase the marks by 9.78(=0.163*60) %.

➤ PART 3-----6 --'1-VARIATE' DESCRIPTIVE ANALYSIS

TABLE NO. 8: 6-'1-VARIATE' ANALYSIS TABLE

MEASURES	VARIABLES					
	M	SS	PE	SL	TR	MU
a.m.	73.992	12.0840	2.8188	29.3344	7.4996	11.8616
s.d.	9.751715	5.864443	2.343154	4.783616	5.137102	5.276803
g1	0.2867417	0.8290903	0.7187482	1.3276170	0.2595679	0.1468140
g2	-1.1798481	0.9734283	-0.3504717	3.4849231	-1.0002420	-1.0486691

1. MEASURE 1-CENTRAL TENDENCY MEASURE-a. m. (ARITHMETIC MEAN)

a) Sem4 % marks-

*On an average the students have scored around 73.992 % of 500,
i.e., 369.96(=73.992 *5) out of 500.*

b) Daily average % duration of self-study

*On an average the students contribute around 12.0840% of 86400 secs,
i.e., 10440.576 secs,
i.e., 174.0096 mins,
i.e., 2.90016 hrs,
i.e., 2hrs & 54mins approximately,
of a day on Self-Study.*

c) Daily average % duration of physical exercise

*On an average the students contribute around 2.8188% of 86400 secs,
i.e., 2435.4432 secs,
i.e., 40.59072 mins,
i.e., 40 mins & 35 secs approximately,
of a day on Physical Exercise.*

d) Daily average % duration of sleep

*On an average the students contribute around 29.3344% of 86400 secs,
i.e., 25344.9216 secs,
i.e., 422.41536 mins,
i.e., 7.040256 hrs,
i.e., 7 hrs & 2 mins approximately,
of a day on Sleep.*

e) Daily average % duration of travel

*On an average the students contribute around 7.4996% of 86400 secs,
i.e., 6479.6544 secs,
i.e., 107.99424 mins,
i.e., 1.799904 hrs,
i.e., 1 hr & mins approximately,
of a day on Travel.*

f) Daily average % duration of media-usage

*On an average the students contribute around 11.8616% of 86400 secs,
i.e., 10248.4224 secs,
i.e., 170.80704 mins,
i.e., 2.846784 hrs,
i.e., 2 hrs & 51mins approximately,
of a day on Media-Usage.*

2. MEASURE 2-DISPERSION MEASURE-s. d. (Standard Deviation)

a) Sem4 % marks-

The s.d. value for Sem4 % marks is 9.751715,
i.e., on an average a student's marks will vary within the range of
[73.992% - 9.751715%, 73.992% + 9.751715%]
i.e., [64.240285%, 83.743715%],
i.e., [321,419] marks on 500.

b) Daily average % duration of self-study

The s.d. value for Daily average % duration of self-study time 5.864443,
i.e., on an average a student's Daily average % duration of self-study time,
will vary within the range of,
[12.0840% - 5.864443%, 12.0840% + 5.864443%],
i.e., [6.219557%, 17.948443%],
i.e., [5373.697248 secs, 15507.45475 secs],
i.e., [89.5616208 mins, 258.4575792 mins],
i.e., [1.49269368 hrs, 4.30762632 hrs],
i.e., [1 hr & 30 mins, 4 hrs & 18 mins] approximately.

c) Daily average % duration of physical exercise

The s.d. value for Daily average % duration of physical exercise time is 2.343154,
i.e., on an average a student's Daily average % duration of physical exercise time,
will vary within the range of,
[2.8188% -2.343154%, 2.8188% + 2.343154%],
i.e., [0.475646%, 5.161954%],
i.e., [410.958144 secs, 4459.928256 secs],
i.e., [6.8493024 mins, 74.3321376 mins],
i.e., [6.8493024 mins, 1.23886896 hrs],
i.e., [7 mins, 1hr & 14 mins] approximately.

d) Daily average % duration of sleep

The s.d. value for Daily average % duration of sleep time is 4.783616,
i.e., on an average a student's Daily average % duration of sleep time,
will vary within the range of,
[29.3344% - 4.783616%, 29.3344% + 4.783616%],
i.e., [24.550784%, 34.118016%],
i.e., [21211.87738 secs, 29477.96582 secs],
i.e., [353.5312896 mins, 491.2994304 mins],
i.e., [5.89218816 hrs, 8.18832384 hrs],
i.e., [5 hrs & 54 mins, 8 hrs & 11 mins] approximately.

e) Daily average % duration of travel

The s.d. value for Daily average % duration of travel time is 5.137102, i.e., on an average a student's Daily average % duration of travel time, will vary within the range of,
 $[7.4996\% - 5.137102\%, 7.4996\% + 5.137102\%]$,
 i.e., $[2.362498\%, 12.636702\%]$,
 i.e., $[2041.198272 \text{ secs}, 10918.11053 \text{ secs}]$,
 i.e., $[34.0199712 \text{ mins}, 181.9685088 \text{ mins}]$,
 i.e., $[34.0199712 \text{ mins}, 3.03280848 \text{ hrs}]$,
 i.e., $[34 \text{ mins}, 3\text{hrs} \& 2 \text{ mins}]$ approximately.

f) Daily average % duration of media-usage

The s.d. value for Daily average % duration of Media-Usage is 5.276803, i.e., on an average a student's Daily average % duration of Media-Usage, will vary within the range of,
 $[11.8616\% - 5.276803\%, 11.8616\% + 5.276803\%]$,
 i.e., $[6.584797\%, 17.138403\%]$,
 i.e., $[5689.264608 \text{ secs}, 14807.58019 \text{ secs}]$,
 i.e., $[94.8210768 \text{ mins}, 246.7930032 \text{ mins}]$,
 i.e., $[1.58035128 \text{ hrs}, 4.11321672 \text{ hrs}]$,
 i.e., $[1 \text{ hr} \& 35 \text{ mins}, 4 \text{ hrs} \& 7 \text{ mins}]$ approximately.

3. MEASURE 3-SKEWNESS MEASURE-g1

a) Sem4 % marks-

The distribution of Sem4 % marks is positively skewed since g1 value (**0.2867417**) is positive.

b) Daily average % duration of self-study

The distribution of Daily average % duration of self-study is positively skewed since g1 value (**0.8290903**) is positive.

c) Daily average % duration of physical exercise

The distribution of Daily average % duration of physical exercise is positively skewed since g1 value (**0.7187482**) is positive.

d) Daily average % duration of sleep

The distribution of Daily average % duration of sleep is positively skewed since g1 value (**1.3276170**) is positive.

e) Daily average % duration of travel

The distribution of Daily average % duration of travel is positively skewed since g1 value (**0.2595679**) is positive.

f) Daily average % duration of media-usage

The distribution of Daily average % duration of media-usage is positively skewed since g1 value (**0.1468140**) is positive.

➤ **NOTE: A positively skewed distribution here implies larger no. of students tend towards lower values of the variable concerned.**

4. MEASURE 4-KURTOSIS MEASURE-g2

a) Sem4 % marks-

The distribution of Sem4 % marks is Platykurtic since g2 value (-1.1798481) is negative, i.e., the frequency distribution concerned is flat, there is not much of peaked-ness, there is not much of concentration around a value, the distribution evenly spreads across the range of [0.100].

b) Daily average % duration of self-study

The distribution of Daily average % duration of self-study is Leptokurtic since g2 value (0.9734283) is positive, i.e., the frequency distribution concerned is not very flat, there is an existence of peaked-ness.

c) Daily average % duration of physical exercise

The distribution of Daily average % duration of physical exercise is Platykurtic since g2 value (-0.3504717) is negative, i.e., the frequency distribution concerned is flat, there is not much of peaked-ness, there is not much of concentration around a value, the distribution evenly spreads across the range of [0.100].

d) Daily average % duration of sleep

The distribution of Daily average % duration of sleep is Leptokurtic since g2 value (3.4849231) is positive, i.e., the frequency distribution concerned is not very flat, there is an existence of peaked-ness.

e) Daily average % duration of travel

The distribution of Daily average % duration of travel is Platykurtic since g2 value (-1.0002420) is negative, i.e., the frequency distribution concerned is flat, there is not much of peaked-ness, there is not much of concentration around a value, the distribution evenly spreads across the range of [0.100].

f) Daily average % duration of media-usage

The distribution of Daily average % duration of media-usage is Platykurtic since g2 value (-1.0486691) is negative, i.e., the frequency distribution concerned is flat, there is not much of peaked-ness, there is not much of concentration around a value, the distribution evenly spreads across the range of [0.100].

5. GRAPHICAL REPRESENTATION-6 HISTOGRAM PLOTS



